Table D.1. Computational efficiency and convergence performance of four coupling strategies.

Configuration	Mean iterations	Wall-clock (min)	Converged steps
State-informed + IQN-ILS	1.981	68.67	100%
State-informed + fixed relaxation	28.671	762.20	8.30%
Linear + fixed relaxation	30.000	919.06	0%
Linear + IQN-ILS	2.001	84.93	100%

Note: Unconverged steps are advanced after reaching the iteration limit of 30 and are included in the mean iteration count.

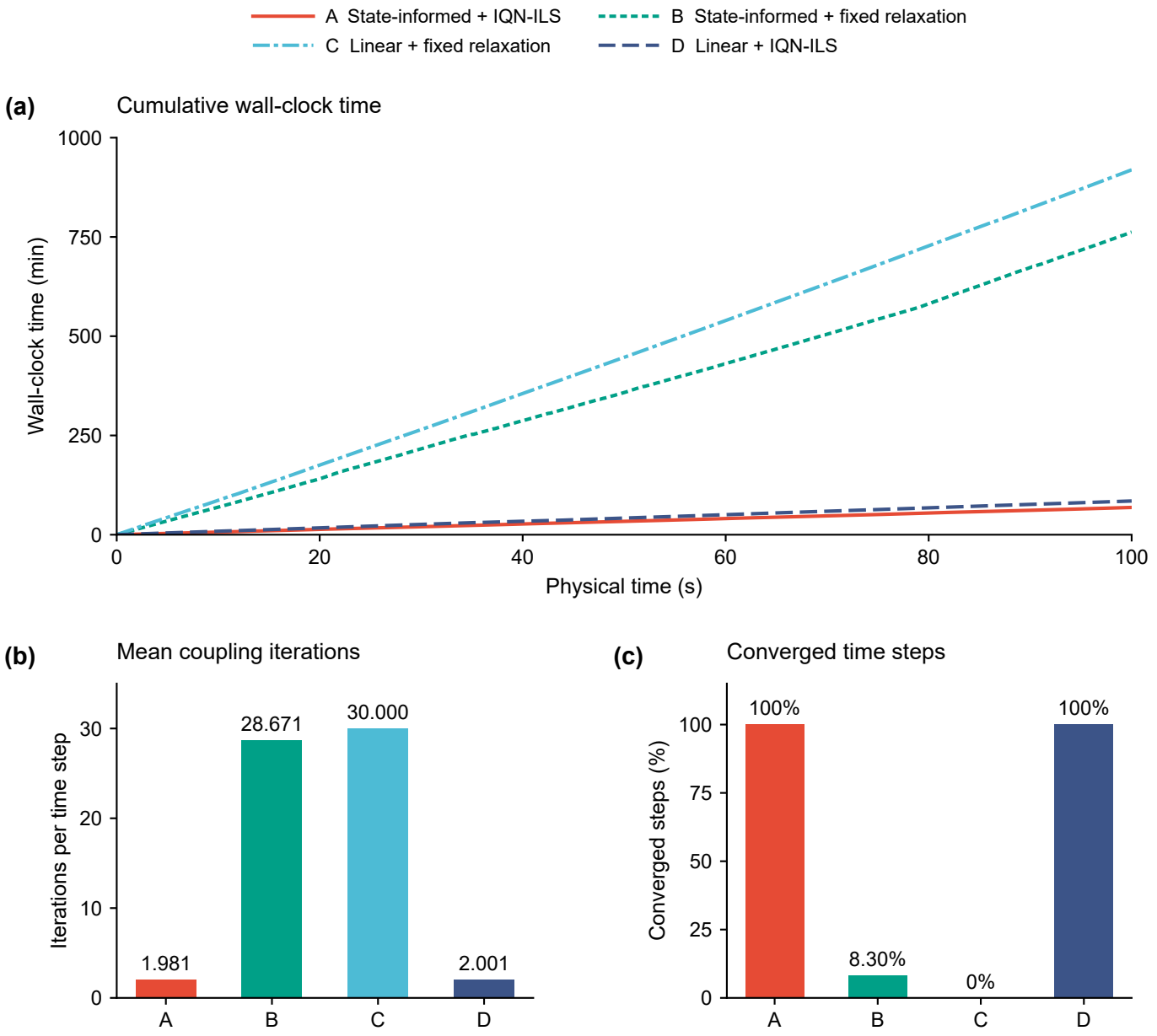


Fig. D.3-1. Computational efficiency and convergence of four coupling strategies.

A maximum of 30 coupling iterations per physical time step is imposed in this comparison. If the convergence criterion remains unsatisfied at this limit, the simulation advances using the final trial state, and the time step is classified as unconverged. This continuation rule allows all four diagnostic runs to cover the same 0–100 s interval. The mean iteration count includes both converged and unconverged steps. The fixed-relaxation runs are therefore used to assess convergence robustness under the prescribed iteration limit, rather than as converged reference solutions.